we are trying to build a simple website app as a prototype for a 86ing pal for frontline dining industry workers. the vision is the app receive reports of missing ingredients from workers through verbal input in natural language, understand which ingredient is out, then update all affected products on various ordering platforms.

The purpose of this prototype only tests the receiving part and act as a tech assessment tool. The technology under inspection is Realtime API, which is an OpenAI technology launched in 2024 that allows users to interact with OpenAI models with real-time audio input. The purpose of this assessment is to evaluate the Realtime model's ability to receive frontline restaurant workers' commands verbally in natural language in a chaotic environment.

The web app will contain a fictional list of ingredients for a Hong Kong style tea restaurant including:
Lemon, Honey, Ribena, Matcha, Waffle, Coffee, Yogurt, Mango Syrup, Lychee Syrup, Peach Syrup, Coconut Milk, Fresh Milk, Oat Milk, Cocoa powder, Blue Curaçao Syrup, Spide, Goofu, Ovaltine, Wonpao. Spide, Goofo, Wonpao are fictional brand specific ingredents.

The prototype will be configured to take user's verbal input, determine if an ingredient is out, and print the unavailable ingredients on screen in text. The prototype will also be configured to tackle certain edge cases via the Prompt feature of Realtime API.

The prototype should be minimalistic. Ideally it should only contain a start conversation button and a dialogue UI with the model. User press a button to start conversation, press again to end. If conversation is idle, terminate the conversation session after 5 seconds to minimize token waste when not used. Prototype should output unavailable ingredient report in this format:
<h2> Got it. The Following Ingredient is out: </h2>

Ingredient 1
Ingredient 2

The prototype is going to be tested on mobile platforms such as phone and tablet so it should be designed mobile first.

I have an openai account and I plan to use gpt-realtime-2.1-mini as the model. I have an api key. when you need it, let me know.

The prototype should not respond casual talkings to save token. Maybe a polite notification that it is only meant to handle ingredient related issues.

Meanwhile, the following scenarios will be tested to simulate real restaurant experience. the prototype should be configured to handle the scenarios represented by those test actions with the Prompt feature of Realtime models.

I have Next.js or Vite in mind for tech stack but feel free to choose whatever tech stack that best suits this use case.

Set 1: Basic Functionality Test

This set aims to test the fundamentals by telling the prototype an ingredient is out. Selected ingredients from all three categories are tested. The tester will tell the prototype the following ingredients are out and check if the prototype reports accurately in order: Lemon, Ribena, Honey, Mango Syrup, Lychee Syrup, Coconut Milk, Blue Curaçao Syrup, Spide, Goofu and Ovaltine. The purpose is to test if the prototype can identify names in different complexity.

Set 2: Irregular Input Test

This set tests how well Reatime models can respond when users call an ingredient in non-standard names. Actions, desired outcome and purpose are listed below.

Action	**Desired Outcome**	**Purpose**
Tell prototype “Blue Curaçao” is out	Report Blue Curaçao Powder is out	Test if missing word affects the result
Tell prototype “Curaçao” is out	Report Blue Curaçao Powder is out	Test if missing word affects the result
Tell prototype “Caocao” is out	Double-check if the user means Cocoa powder or Curaçao. Should not assume.	Test if the model takes a misspelt word for granted
Tell Prototype “Milk” is out	Double-check if user means Coconut milk, Fresh milk or Oat milk	Test if the model can respond to ambiguous inputs
Tell Prototype 3 random ingredients are out, with a very short break between the 3 statements	Accurately list the 3 ingredients without mistakes or missing anything	Test if the model will break under consistent input, simulating the extreme situation where multiple employees report an outage in a row

Tell prototype 5 random ingredients are out	Accurately list the 5 ingredients without mistakes or missing anything	Simulates the situation where the same employee reports multiple outages in one sitting.

Set 3: Disrupted Input

This set tests how Realtime model react when input is disrupted in various ways in chaotic restaurant setting.

Action	**Desired Outcome**	**Purpose**
Tell prototype “we ran out of” and cut from there	Prototype provides a message saying you are cut off without doing anything	Test if prototype can handle broken sentences
Tell Prototype “We ran out of…”, pretend tester is interrupted by other things, wait for 1-2 seconds, then com back to finish the sentence “...Ribena”	Prototype either provides a message saying you are cut off without doing anything or register the ingredient accurately	Test if prototype can handle broken sentences
Tell Prototype “we ran out of Lemon”. Move prototype away from sound source when the sentence is half finished	Prototype accurately reports lemon is out (desired) or ask for confirmation without doing anything (acceptable)	Simulate the situation where workers are too busy, that they are running to the next task before finishing the sentence
Tell Prototype “we ran out of Coconut Milk”. Move prototype away from sound source when the sentence is half finished	Prototype accurately reports coconut milk is out (desired) or ask for confirmation without doing anything (acceptable)	same as above
Tell Prototype “We ran out of honey” while another testee says “No Mango Syrup?” loudly in ambient	Prototype accurately reports honey is out (desired), or asks for confirmation (acceptable).	Test if prototype gets confused by ambient distractions